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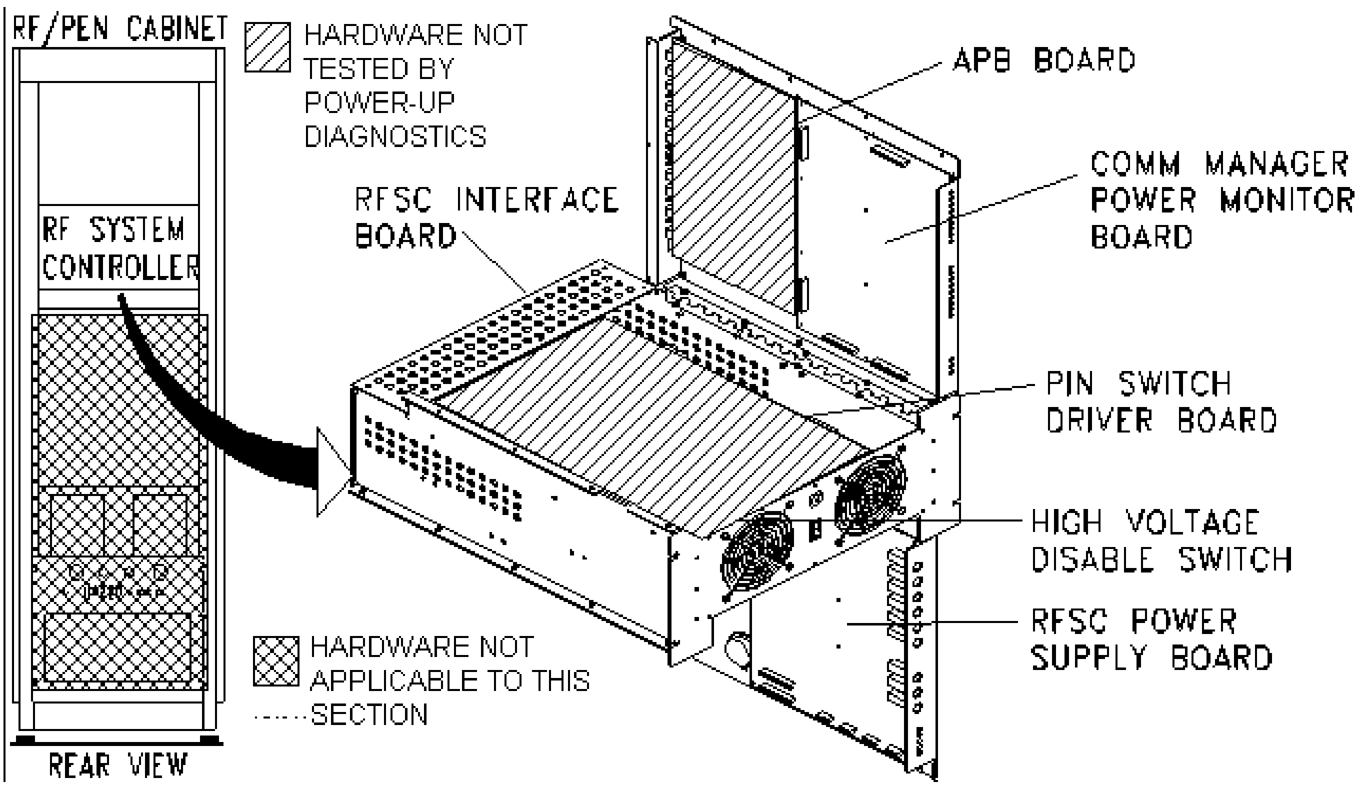
Description - This procedure explains how to start power monitor tests, what test LEDs are provided, where error messages are logged, and test descriptions for the RF/PEN Cabinet.

1- HARDWARE TESTED

Power Monitor Power-up/Confidence diagnostics are available for:

- Communication Manager/Power Monitor Board
- ±15V Power Supplies

Refer to Illustration L1608b for hardware locations.



POWER MONITOR TESTED HARDWARE LOCATION
ILLUSTRATION L1608B

2- STARTING POWER MONITOR TESTS

Only EPROM-based tests are available. They are executed when power is first applied to the RF System Controller. Since the operating system is not yet loaded, all errors are reported only on local test indicators.

Start the EPROM-based tests as follows:

1. Turn RF/PEN Cabinet power on (MR1A14CB6).

2. If power is already on, turn off and then back on.

3- POWER MONITOR TEST INDICATORS

The main source for reporting errors is the red Fault LED located on RF System Controller Module Front Panel. At the beginning of test execution, all LEDs are turned on (lamp test). Typically, if any test error is encountered, further testing halts, the yellow and green LEDs are turned off (except the top green LED stays on) and the red LED blinks. In the case, however, where testing doesn't get far enough before an error occurs, a combination of red/yellow/green LEDs might be illuminated. If all tests complete without errors (testing takes 0.5 seconds), all red and yellow LEDs go out and the two green LEDs remain on.

Additional power-up test error reporting is also provided by the RF System Controller Module Front Panel LEDs (refer to Table 1). The red Fault LED will be blinking also.

TABLE 1
DECODING POWER MONITOR TEST LEDS

Head	Surface	Transmit	Receive	Failed Test
ON	OFF	ON	OFF	POWER SUPPLY
ON	ON	OFF	OFF	PROM CHECKSUM
ON	ON	OFF	ON	RAM
ON	ON	ON	OFF	CPU TIMER
ON	ON	ON	ON	CPU COMMUNICATIONS (Serial Port)

4- POWER MONITOR TEST DESCRIPTIONS

Refer to Table 2 for Communication Manager/Power Monitor Board tests.

The Power Supply test verifies proper voltage output from the ±15VDC analog power supplies.

TABLE 2
CONTROL BOARD TEST DESCRIPTIONS

CPU Test - The purpose of this test is to verify operation of the 8031 CPU; it includes testing of the status flags, timers, and serial port.

EPROM Test - A checksum test is performed on the EPROM device to verify the integrity of its data (i.e., data hasn't changed since programmed). The computed checksum is verified against the embedded checksum. This checksum is different for 1.5T and 1.0T systems.

RAM Test - Test time is ~ 15 seconds. This test verifies the 4MB of CPU Board DRAM as follows:

- Memory Cell Test - This test verifies each memory bit can be set and cleared. Patterns used are: 55H and AAH.
- Address Test - This test verifies that all address lines are functioning properly and all memory locations are addressable. Each memory location is written with the hex value equal to its memory address plus a "page" offset, where each "page" contains 256 bytes. Each location is then read back and verified for correct data.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	July 14, 1998	J. Saperstein	Initial conversion from Toolbook to Word.